

processor for the desktop segment that is currently being ruled by dual- and quad-core processors. This processor might be launched sometime in March this year.

The processor, codenamed Gultown, will use Intel's 32-nanometre (nm) fabrication technology and Westmere architecture, which means it will also feature some of the interesting power-saving techniques used in Intel's 45nm products. The power management functions of this architecture will ensure power savings by limiting the power consumed by the idle cores not being utilised by the applications running at any point of time.

Apart from that, the chip has a larger data cache—12 MB of L3 compared to the 4 MB present in its dual-core counterparts. As John Morris and Sean Portnoy observe on their famous blog on laptops and desktops [blogs.zdnet.com/computers], this data cache, when combined with the extra cores, would result in a chip that is larger and contains 1.17 billion transistors. Intel says that it uses some of those extra transistors to speed up tasks like data encryption and decryption.

The six-core processor will also be available in a dual-socket server version.

AMD too plans to launch its six-core desktop CPU this year. Codenamed Thuban, this processor will pack six cores and a DDR3 memory controller onto a 45nm die. It is believed that Thuban will be backwards-compatible with existing AM3 and AM2+ motherboards, which gives a flexible upgrade path to existing AMD users.

Six more for the servers

If six cores is the current high score for desktop CPUs, server-focused processors can boast of six more.

AMD's new server processor, codenamed Magny Cours, is designed for two- and four-socket servers and uses faster DDR3 memory. This processor will be offered in the market as Opteron G100, in both eight- and 12-core versions. Both chips will be manufactured using a 45nm manufacturing process. The company has also hinted that by 2011, Magny Cours will be replaced with Interlagos—a more powerful 32nm chip that will come with 12 and 16 cores based on AMD's Bulldozer microarchitecture.

The company has also revealed a new Opteron chip (codenamed Lisbon) for servers with one or two processor sockets. This will be available in four- and six-core versions as the Opteron 4100 series. By 2011, Lisbon will be succeeded by Valencia—a 32nm chip offering six and eight cores based on Bulldozer.

A team with multiple capabilities

"Last year, we launched the six-core Istanbul processor; we are currently working on the eight-core and 12-core processors, and plan to launch them by the first half of 2010. Apart from this, we are also planning a six-core desktop CPU. While these homogeneous multi-core processors will continue to evolve over the next couple of years, you will see a movement towards heterogeneous multi-core processors in the future," comments Vamsi Krishna, senior manager, Technical.

Coming up...

1. Six-core processors for desktops
2. Server processors with eight, twelve and even more cores
3. Heterogeneous multi-core processors
4. Smaller, more powerful multi-core chips based on 32nm logic technology
5. Single-chip cloud computers
6. Scalable multi-core architectures
7. Higher-bandwidth I/O and communications, for improved performance of multi-core chips
8. Better parallel programming tools, model-based tera-scale applications and thread-aware execution environments, to make better use of multi-core hardware

AMD: "AMD has plans to release its first heterogeneous multi-core processor (codenamed Fusion), where we integrate multiple x86 cores and graphical processing unit (GPU) cores on a single die. This will be the beginning of a new multi-core revolution in the future."

Deep down, the idea behind Fusion is to enable a mix-and-match strategy at the silicon level, through a broad range of design initiatives. AMD will provide a range of application-specific cores that can easily be combined and heaped onto a processor die and fabricated at a low cost. A quad-core processor might contain different combinations of cores, say, two general-purpose cores and two specialised processor cores, or one general-purpose core and three specialised cores, and so on. This implies that a processor can be put together from heterogeneous cores, based on the end-use and workload.

Krishna adds, "The Fusion project completely leverages the multi-core concept and Direct Connect architecture, enables a homogeneous programming model for all AMD products, and standardises the coprocessor interface for on-die and platform connectivity. Fusion-based processors, with the CPU and GPU integrated in a single architecture, should make the life of software programmers and application developers much easier."

Cloud on a chip

System-on-chips are old news. Now, here is a whole cloud on a chip—Intel's brainchild with 48 cores.

"Most recently, we announced the prototype, single-chip cloud computer (SCC). This research chip contains the most Intel architecture cores ever integrated on a silicon CPU chip—48 cores. It incorporates technologies intended to scale multi-core processors to 100 cores and beyond, while consuming less electricity than two standard household light bulbs," says Vasantha Erraguntla, engineering manager, Intel India.

"Architecturally, the chip resembles a small cluster or 'cloud' of computers on a chip. It was designed as a concept vehicle for parallel software research. We will be working with industry research and academia partners to further parallel computing research using this vehicle."

You might recollect that the same team also successfully tested an 80-core teraflop processor a few years ago. Following that, the team was confident of the success of the